Predicting Recovery Time After Climbing Injuries

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Abstract

We investigate factors associated with recovery time after rock climbing injuries among recreational climbers. Using a cross-sectional survey of 237 climbers from Lum and Park (2019), we model recovery as a three-category outcome—short (<6 months), medium (6–24 months), and long (>2 years or still recovering)—using multinomial logistic regression. Predictors include demographics, climbing experience and grades, injury location, and treatment choices such as surgery and physical therapy. The model achieves about 60% accuracy on a held-out test set and performs best for short recovery, but has limited ability to distinguish medium and long recoveries, reflecting class imbalance and overlapping feature patterns. Coefficient magnitudes highlight injury location, climbing experience and difficulty, and surgery status as key predictors. We note limitations such as small sample sizes for long recoveries and unmeasured injury severity, and suggest future work using richer data, continuous-time outcomes, and more flexible models to improve prediction.

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1. Introduction

In recent years, rock climbing has shifted from a niche activity to a popular recreational sport, with more indoor climbing gyms, greater media coverage, and its debut as an Olympic event in 2020. As recreational climbers and members of the outdoor community, we notice how common injuries are in everyday climbing sessions—not just among elite athletes. Climbers frequently deal with falls, finger and shoulder problems, and overuse injuries. These observations naturally lead to our central question: **which factors contribute to recovery time after a climbing-related injury?**

Compared with many traditional sports, climbing injuries and recovery have not been studied as deeply, and there is relatively little work on predicting how long it takes climbers to return after an injury. At the same time, many recreational climbers are relatively young (13–35 years old) and often lack strong financial resources or access to high-quality medical care, so they frequently rely on age, “good genetics,” or informal advice instead of structured rehabilitation. These gaps in both research and support make it especially important to understand which factors are linked to shorter or longer recovery times, so climbers can make safer and more informed decisions.

Through this project, our group hopes to help make climbing a safer and better-understood sport by clarifying which factors affect recovery time after injury. We build a multinomial logistic regression model to predict whether a climber’s recovery will be short, medium, or long, and to identify which factors—such as injury location, experience level, and surgery status—are most strongly associated with these outcomes. By evaluating both the model’s performance and the importance of individual

predictors, we aim to provide insights that can help everyday climbers make more informed decisions about training, injury management, and recovery, and to contribute to the growing body of research on rock climbing injuries.

2. Dataset and Preprocessing

2.1 Data source

The data for this project come from a cross-sectional survey of recreational climbers collected by Lum and Park (2019), as part of a study on rock climbing injuries and time to return to sport. The survey was administered as an anonymous questionnaire, approved by the UC Davis Institutional Review Board (IRB), and the resulting dataset is publicly available on Mendeley Data. Using this high-quality dataset, we examine how injury characteristics, climber demographics and experience, and treatment choices (such as surgery or physical therapy) relate to the recovery-time category.

2.2 Cleaning steps

Starting from the raw survey file, we applied several data-cleaning steps consistent with our presentation. First, for sample selection, we kept only rows corresponding to real respondents and removed non-data rows such as extra headers and totals, resulting in a final sample of **237 climbers**. Next, we removed identifiers and unrelated fields by dropping respondent IDs, timestamps (start and end dates), IP addresses, emails, names, and any other personal information. This step both protects participant privacy

and focuses the analysis on variables relevant to recovery. We then selected variables that are likely to be related to recovery duration, including demographics (age group, gender), climbing experience (years of climbing), performance level (rope and boulder grades), injury characteristics (injury location and type), treatment variables (whether the climber had surgery, whether they were satisfied with surgery, and whether they had physical therapy), and the recovery-time category, which serves as our outcome. Finally, because many of these variables are categorical (for example, age ranges, grade ranges, injury locations, and surgery options), we encoded them as 0/1 dummy variables for modeling. For instance, the injury location variable was expanded into indicators such as injury_shoulder and injury_spine, and rope grade was represented with indicators such as route_511a_511d and route_512a_512d.

2.3 Outcome variable

Recovery time in the original survey was reported in several ordered categories, including less than 1 month, 1–2 months, 2–6 months, 6–12 months, 1–2 years, more than 2 years, and “never fully recovered / still recovering.” These categories describe how long it took (or is still taking) for climbers to return to their sport after an injury.

For our analysis, we grouped these responses into three broader recovery categories: Short (less than 6 months), Medium (6–24 months), and Long (more than 2 years, never fully recovered, or still recovering). This three-level outcome serves as the dependent variable in our model and is analyzed using multinomial logistic regression.

3. Exploratory Data Analysis

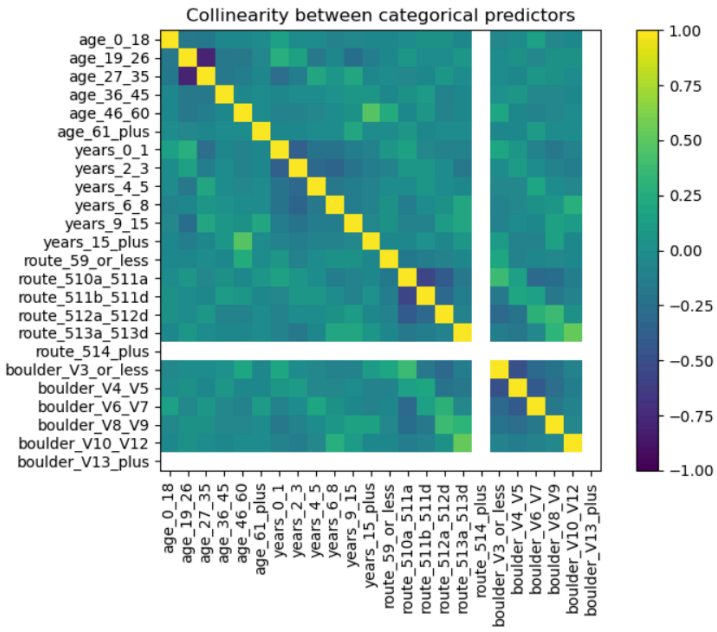
3.1 Recovery-time distribution

Bar plots of the recovery-time categories show that most climbers recover within 2–6 months, and the majority are back within less than 1 year, but there is a noticeable minority who need 1–2 years or longer, and some report that they never fully recover or are still recovering. Overall, the recovery groups are imbalanced, with many more short recoveries than long ones, which is important because this imbalance can make it harder for models to learn reliable patterns for the rarer medium and long categories.

3.2 Recovery by treatment type

Treatment approach is a key factor that can directly influence how quickly and fully a climber returns to their prior level after an injury. We therefore examined recovery times grouped by treatment categories: no surgery and no physical therapy, physical therapy only, surgery with satisfaction, and surgery without satisfaction. The bar plots suggest that the mix of fast versus slow recovery changes across these treatment types, with some surgery groups showing a higher proportion of long-term or still-recovering cases. This pattern indicates that treatment type may be an important predictor and should be included in our models, although we must also keep in mind that treatment choice is not randomized and may be confounded by injury severity.

3.3 Collinearity among predictors



We computed a correlation heatmap for the one-hot encoded predictors based on age group, years of climbing, rope grade, and boulder grade. Because each climber can only be in one category within a group (for example, one age range or one rope-grade range), the dummy variables inside each group are very strongly related and often show strong negative correlations—for instance, if `age_19_26 = 1`, all other age dummies must be 0. This strong pattern also reflects that most of our sample is 19–26, so our results mainly represent younger climbers. In contrast, correlations across different groups (such as age vs. years climbing or years vs. grade) are much weaker, indicating that these predictors capture different aspects of the climber. Because of the strong within-group relationships, we dropped one dummy in each categorical group to serve as a reference level in the logistic regression model, which avoids multicollinearity and makes the coefficient estimates easier to interpret.

4. Modeling Approach

We used a multinomial logistic regression model with recovery group as the outcome, where “short” recovery is defined as less than 6 months, “medium” as 6–24 months, and “long” as more than 2 years, never recovered, or still recovering. The predictors in the model include indicators for age group, years of climbing, rope and boulder grades, injury location (for example, pelvis/hip, spine, hand/elbow, shoulder), and treatment-related variables such as surgery and physical therapy (including categories like no surgery, surgery with satisfaction, and surgery with dissatisfaction). We split the data into 80% training and 20% test sets, stratified by the recovery group. The model was trained on the training set and evaluated on the held-out test set using overall accuracy, class-wise precision, recall, F1 scores, confusion matrices, and ROC curves for each recovery category.

5. Results

5.1 Overall accuracy

On the test set of 47 climbers, the multinomial logistic regression model achieved an accuracy of about 60%, correctly predicting 20 out of 47 recovery outcomes. This is only slightly better than a simple majority-class baseline that always predicts “short” recovery (about 28 out of 47 short cases, or ~60%), but the model has the advantage of

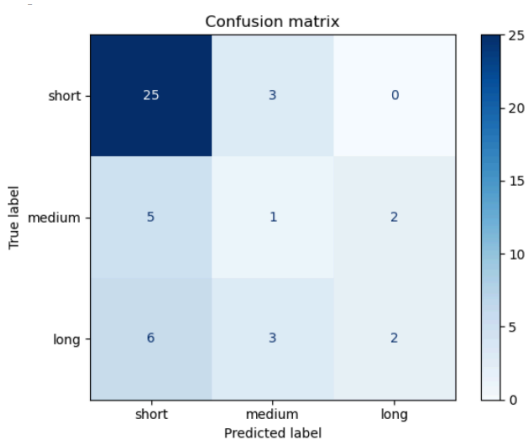
sometimes correctly identifying medium and long recoveries, which the baseline never would.

```
=== Logistic Regression (3-group: short / medium / long) ===
Accuracy: 0.5957446808510638
F1 (macro): 0.39375
F1 (weighted): 0.550531914893617

Classification report:
              precision    recall  f1-score   support

       short       0.69      0.89      0.78        28
      medium       0.14      0.12      0.13         8
        long       0.50      0.18      0.27        11

    accuracy                           0.60        47
   macro avg       0.45      0.40      0.39        47
weighted avg       0.56      0.60      0.55        47
```

The classification report shows that performance is best for the short group: precision is 0.69 and recall is 0.89, meaning that most of the climbers the model labels as short actually are short, and it captures most of the true short-recovery climbers. However, performance is much weaker for the medium and long groups. For medium recovery, precision (0.14) and recall (0.12) are very low, so the model rarely predicts medium and is often wrong when it does. For long recovery, precision is moderate (0.50), but recall is only 0.18, meaning it correctly identifies only a small fraction of true long-recovery cases. The macro F1 score (0.39) summarizes this imbalance by averaging performance across all three classes, while the higher weighted F1 score (0.55) is driven mostly by the good performance on the majority short group. Overall, the model is reasonably good at detecting short recoveries but struggles to reliably separate medium and long recoveries, reflecting both the class imbalance and the difficulty of the prediction task.

5.2 Confusion matrix and F1-scores

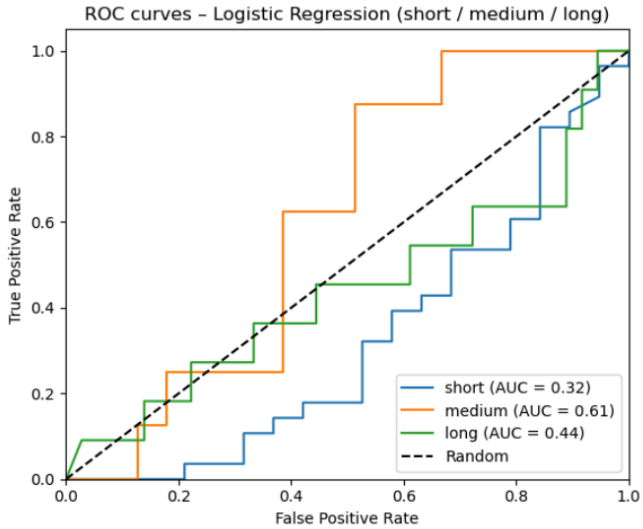
From the confusion matrix:



For short recovery, the model correctly predicted 25 climbers as short, misclassified 3 as medium, and none as long, giving precision of about 0.69, recall of about 0.89, and an F1 score around 0.78. For medium recovery, only 1 climber was correctly predicted as medium, while 5 were misclassified as short and 2 as long, resulting in a very low F1 score of about 0.13. For long recovery, the model correctly identified 2 climbers as long, but misclassified 6 as short and 3 as medium, with an F1 score of about 0.27. Overall, these results show that the model performs well for short recovery but struggles to distinguish between medium and long, often “collapsing” many medium and long cases into the short category, likely because the short group is much larger and easier for the model to learn.

5.3 ROC curves

```
AUC by class:
short: AUC = 0.324
medium: AUC = 0.606
long: AUC = 0.443
```

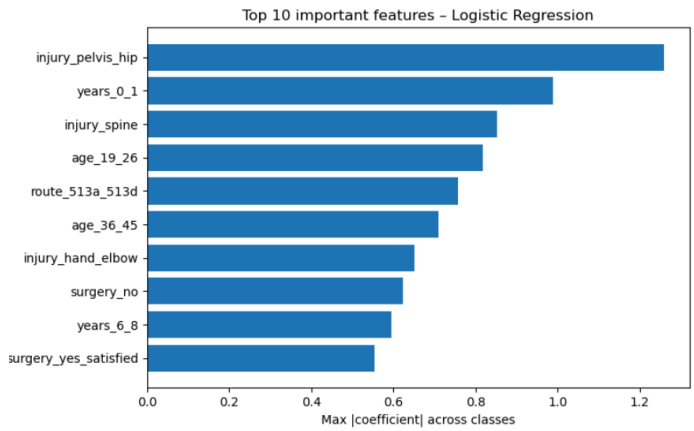


We also plotted ROC curves for each class using a one-vs-rest approach. For the short recovery group, the AUC is about 0.32, which is worse than random and suggests that the model does a poor job separating short from non-short cases based on predicted probabilities. For the medium group, the AUC is about 0.61, slightly better than random, indicating that the model has some ability to identify medium recoveries. For the long group, the AUC is about 0.44, which is close to random guessing. Overall, these ROC results are consistent with the confusion matrix and F1-scores: the model has limited discriminative power, especially for the short and long recovery categories.

6. Feature Importance

To interpret the model, we examined the absolute size of the logistic regression coefficients. A bar chart ranks features by the magnitude of their coefficients, with larger bars indicating a stronger impact on the predicted recovery group.

	max_abs_coef
injury_pelvis_hip	1.258701
years_0_1	0.989528
injury_spine	0.851388
age_19_26	0.817201
route_513a_513d	0.757693
age_36_45	0.709050
injury_hand_elbow	0.650570
surgery_no	0.623494
years_6_8	0.594054
surgery_yes_satisfied	0.554251
injury_shoulder	0.514481
boulder_V4_V5	0.450899
surgery_yes_not_satisfied	0.423376
route_512a_512d	0.394948
age_61_plus	0.384441



Injury location is one of the most important factors. Variables such as injury_pelvis_hip, injury_spine, injury_hand_elbow, and injury_shoulder have some of the largest coefficients, suggesting that *where* a climber is injured is closely linked to whether they have a short, medium, or long recovery.

Climbing experience also matters. Experience levels like years_0_1 and years_6_8 show notable effects, indicating that both very new climbers and those with several years of experience can have different recovery patterns compared with other groups.

Age group plays a role as well. Categories such as age_19_26, age_36_45, and age_61_plus appear important, suggesting that recovery profiles differ across life

stages and that younger and older climbers may face different risks or recovery challenges.

Difficulty level is another influential set of predictors. Higher rope grades (for example, route_513a_513d and route_512a_512d) and intermediate boulder grades like boulder_V4_V5 are associated with distinct recovery profiles, which may reflect differences in training load, movement demands, or injury mechanisms at different difficulty levels.

Finally, treatment indicators have strong coefficients. Variables such as surgery_no, surgery_yes_satisfied, and surgery_yes_not_satisfied show that whether a climber has surgery, and how satisfied they are with it, is closely tied to recovery time.

Overall, the coefficient patterns highlight injury location, climber experience and age, climbing difficulty, and surgery status as the main drivers of the predicted recovery category.

7. Discussion and Limitations

Our analysis shows that a simple multinomial logistic regression can capture some patterns in climbing-injury recovery, but its predictive power is limited. The model performs relatively well for short recovery, which is the majority class, but performance for medium and long recovery is much weaker, with low F1-scores and AUC values close to 0.5.

A major reason for this is **class imbalance**: short recoveries are much more common than long ones, so the model has many more examples to learn from. Medium and long categories have

far fewer observations, which makes them harder to model reliably. In addition, there are likely **overlapping features**—some medium and long cases may look very similar to short cases in terms of age, experience, grades, and treatment. When classes overlap this way, a linear model like logistic regression has trouble drawing clear boundaries.

There are also **unmeasured confounders**. Recovery time can depend on factors not captured in the survey, such as exact injury severity, quality and timing of medical care, adherence to rehab protocols, and individual physiology. Finally, **model simplicity** is a limitation: multinomial logistic regression assumes a linear relationship in the log-odds space, while real recovery patterns may be more complex and non-linear than this model can represent.

8. Future Work

To improve prediction and understanding of recovery time, several extensions are possible. First, collecting more data—especially for medium and long recoveries—would help reduce class imbalance and give the model more examples to learn from. Instead of using broad categories, recovery could be modeled as continuous time (for example, number of weeks), using survival analysis or regression models. More flexible modeling approaches, such as tree-based ensembles (random forests or gradient boosting) or regularized neural networks, could also be explored to better capture non-linear interactions between injury type, experience, and treatment. In addition, resampling or re-weighting techniques, such as class weights or SMOTE, could be used to address class imbalance and potentially improve performance on the rarer medium and long recovery groups.

9. Conclusion

We cleaned and analyzed a climbing-injury survey of **237 recreational climbers**, creating a dataset where recovery time was grouped into **short, medium, and long** categories. Using a multinomial logistic regression model, we obtained around **60% test accuracy**, with strong performance for short recovery but weak discrimination between medium and long recoveries.

Across multiple analyses—EDA, model coefficients, and feature importance—we consistently found that **injury location**, **climbing experience and grade**, and **surgery status** are key factors associated with recovery time. However, due to class imbalance, limited sample size, and the complexity of recovery processes, the current model has limited predictive power, especially for long-term or incomplete recoveries.

This project provides an initial statistical view on climbing-injury recovery but also highlights the need for richer data and more flexible methods to better support climbers and clinicians in understanding and predicting recovery outcomes.

References

Lum, Z. C., & Park, L. (n.d.). *Rock climbing injuries and time to return to sport in the recreational climber.* ScienceDirect.

Lum, Z. (2019, April 17). *Data for: Rock climbing injuries and time to return to sport in the recreational climber.* Mendeley Data.